Takanao Tanaka

UC Berkeley
Agricultural and Resource Economics

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Education

2021 –	PhD in Agricultural and Resource Economics, UC Berkeley
2019 – 2021	MPhil in Social Science, Hong Kong University of Science and Technology
2014 – 2016	M.A. in Public Policy, U Tokyo & National U of Singapore
2009 – 2014	B.A. in Economics, Keio University

Field Interest

Energy and Environmental Economics, Public Economics, Industrial Organization

References

Joseph Shapiro
Agricultural & Resource Economics
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James Sallee
Agricultural & Resource Economics
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Reed Walker
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Job Market Paper

Electricity Supply Shocks and Manufacturing: Evidence from the Fukushima Nuclear Accident (with Daiji Kawaguchi) Available here

We show that adverse shocks to electricity supply—an input accounting for only 2% of revenue—can cause substantial and persistent declines in industrial production. Using confidential establishment-level data and instrumenting with initial supply conditions, we find that shocks following the 2011 Fukushima nuclear accident reduced Japanese manufacturing electricity use by 6.4% and output by 1.8%, implying annual output losses of 5.3 trillion yen (US$44 billion). Establishments adjusted little through energy substitution, product switching, outsourcing, within-firm reallocation, or relocation. Our production function highlights two mechanisms that amplify output losses: establishments cannot easily substitute other inputs for electricity, and reduced electricity use leaves machinery and equipment temporarily idle. We quantify how electricity-market reforms, such as regional market integration, can mitigate these losses.

Working Papers

Combating Noise Pollution: Policy, Exposure, and Welfare (with Takeru Sugasawa and Shinsaku Takikawa) Available here

This paper provides the first evidence on how national noise-control policies affect noise exposure and welfare. We assemble two decades of largely untapped nationwide measurements of road-traffic noise from over 14,000 locations across Japan and detailed data on noise-mitigation policies obtained through freedom-of-information requests. From 2002 to 2022, roadside noise fell by 2.8 decibels, equivalent to roughly a 20% decline in perceived loudness. We attribute about 75% of this decline to noise-control policies, particularly tighter vehicle noise-emission standards and the expansion of low-noise pavement. Residents are willing to pay 1.42% of residential land value for a one-decibel reduction in road-traffic noise. We estimate that Japan's national noise-control policies generated approximately 6.8 trillion yen (US$57 billion) in benefits over two decades. These benefits suggest substantial unrealized gains in countries with weaker noise-control policies, such as the United States.

Authoritarian Control through the Judiciary Available here

Authoritarian governments may use courts to punish groups perceived as political threats. We examine this using 6.4 million Chinese criminal verdicts. Ethnic minorities receive 6.8% longer sentences and face 6.5% and 14.0% higher probabilities of life imprisonment and the death penalty, respectively. These additional penalties are particularly severe for crimes associated with social unrest, for less assimilated ethnic groups, and in regions experiencing heightened social tensions. Penalties against minorities intensified in minority-dominant prefectures during Uyghur unrest and when local political leaders sought promotions. The disparities, however, declined over time and almost disappeared in 2021; we discuss the potential causes.

Publications

1. Air Pollution and the Airborne Diseases: Evidence from China and Japan (with Guojun He and Yuhang Pan) ***Journal of Environmental Economics and Management***, 2025 Link
2. Energy Saving May Kill: Evidence from the Fukushima Nuclear Accident (with Guojun He) ***American Economic Journal: Applied Economics***, 2023 Link
3. Short and Medium-Term Impacts of Strict Anti-Contagion Policies on Non-COVID-19 Mortality in China (with Qi J, D Zhang, et al.) ***Nature Human Behaviour***, 2022 Link
4. Increase in suicide following an initial decline during the COVID-19 pandemic in Japan (with Shohei Okamoto) ***Nature Human Behaviour***, 2021 Link
5. The Short-term Impacts of COVID-19 Lockdown on Urban Air Pollution in China (with Guojun He and Yuhang Pan) ***Nature Sustainability***, 2020 Link

Works in Progress

1. Environmental Regulation, Cost, Price and Welfare (*Census project approved*)

Fellowships, Awards, and Grants

2026 – 2028	Environmental Research Grant, Sumitomo Foundation, approx. $31,600.
2024 – 2026	Project Research Program Grant, IER Hitotsubashi University, $7000.
2022 – 2025	Japanese Student Services Organization (JASSO) Graduate Student Fellowship.
2022	17th Applied Econometrics Conference, Best Paper Award.
2020	15th Applied Econometrics Conference, Excellent Paper Award
2020 – 2021	2021 Annual iHEA Student Paper Prize, Second Prize.
2020 – 2021	Japan Center for Economic Research, Co-investigator, $9000.
2019 – 2021	Scholarship from Bai Xian Asia Institute.
2015 – 2016	Scholarship from Li Ka Shing Foundation.

Presentations

2026–27: ASSA Annual Meeting (D.C.); NBER Japan Project Meeting; Camp Resources; UC Berkeley (EERE)

2025–26: Institutional Barriers to Climate Adaptation (Stanford); LSE Environmental Camp (LSE); IPWSD (Columbia); UC Berkeley (EERE)

2021–24: AERE Annual Conference; JADE-JTREE Workshop (GRIPS); Osaka; SWET; Tsukuba; Occasional Workshop (UCSB); UC Berkeley (EERE)

Services

Referee Service: *American Economic Journal: Economic Policy, Journal of Environmental Economics and Management, Journal of Health Economics, Journal of Economic Behavior & Organization, China Economic Review, Economic Inquiry, Economic Modelling, Nature Climate Change, Scientific Reports*

Workshop Organizing: Student Organizer, EREE Workshop, UC Berkeley, 2024–25

Teaching Experience

UC Berkeley: GSIs for Environmental Economics, Intermediate Microeconomics (Undergrad)
HKUST: TAs for Research Methods in Social Science, Environmental Economics (Undergrad)

Professional Employment

2016 – 2018 Consultant, CSR Unit, Deloitte Consulting, Tokyo, Japan.